

GrounUp Enterprise Specification

Phase 21 - Master Enterprise Consolidation, Traceability & Implementation Baseline

Version 1.0.0 / Review-ready / Proposed lock: GES-P21-v1.0.0

1. Mission

Consolidate the complete enterprise specification into one governed implementation baseline that human developers and AI coding systems can follow without losing source-phase requirements.

2. Source Baseline

This build inventories 20 Phase 01-20 ZIP repositories totaling 866 governed source files. Phase 00 remains the product-definition/requirements-governance source required in final approval.

3. Consolidated Registers

The source consolidation contains 4785 requirements, 4775 tests, 2143 entities, 607 engines, 794 API records, 494 UI records and 4517 source traceability rows.

4. Reconciliation Policy

Machine checks identify structural issues such as duplicate IDs, exact normalized duplicates and missing source mappings. They do not silently resolve semantic conflicts. Business, architecture, security, data and AI-governance conflicts require owner disposition.

5. Build Order

The implementation baseline preserves the approved dependency waves from product/governance and shared platform foundations through domain systems, field intelligence, scheduling/commercial controls, production convergence and final lock.

6. Developer & AI Handoff

Every implementation item should cite source requirement IDs and mapped tests. Missing rules become governed gaps rather than assumptions. Approved schemas, permissions, audit controls, tenant boundaries and release gates are compatibility contracts.

7. Enterprise Lock

GES v1.0 becomes Locked only after blocking findings are closed or formally excepted and required Product, Architecture, Engineering, Data, Security, AI Governance, QA, Operations and Business/UAT approvals are recorded.

8. Implementation Waves

Wave 0 - Phase 00-01

Product definition, requirements governance, decision rights and lock discipline. Exit gate: Approve scope and governance.

Wave 1 - Phase 02-04

Core platform, enterprise data model and master library registry. Exit gate: Stabilize shared schemas, tenancy, identity and libraries.

Wave 2 - Phase 05-06

Estimating/cost intelligence and AI construction intelligence. Exit gate: Deliver core construction intelligence.

Wave 3 - Phase 07-12

Project management, CRM, HR, finance, procurement and fleet. Exit gate: Build operational business platform.

Wave 4 - Phase 13-17

BI, HSE, documents/drawings, GIS/reality capture, machine control. Exit gate: Build intelligence, compliance and field/digital systems.

Wave 5 - Phase 18-19

Scheduling/resource planning and commercial/contracts/claims. Exit gate: Integrate time, resources and commercial control.

Wave 6 - Phase 20

Enterprise integration, deployment, testing, governance and production readiness. Exit gate: Converge and harden platform for production.

Wave 7 - Phase 21

Master consolidation, cross-phase reconciliation, final traceability and implementation baseline. Exit gate: Approve and lock GES v1.0 implementation baseline.

9. Consolidated Counts

Artifact	Count
Source Phases	20
Source Files	866
Consolidated Requirements	4785
Consolidated Tests	4775
Consolidated Entities	2143
Consolidated Engines	607
Consolidated Apis	794
Consolidated Ui Surfaces	494
Consolidated Traceability Rows	4517
Phase21 Requirements	180
Phase21 Tests	180
Structural Findings	278

10. Phase Catalog

Phase 01 - Governance: 25 requirements, 15 tests, 23 entities, 0 engines, 16 APIs, 12 UI surfaces.

Phase 02 - Core Platform: 60 requirements, 60 tests, 36 entities, 12 engines, 22 APIs, 12 UI surfaces.

Phase 03 - Enterprise Data Model: 75 requirements, 75 tests, 82 entities, 10 engines, 12 APIs, 8 UI surfaces.

Phase 04 - Master Library Registry: 97 requirements, 97 tests, 0 entities, 12 engines, 24 APIs, 0 UI surfaces.

Phase 05 - Estimating Cost Intelligence: 108 requirements, 108 tests, 40 entities, 20 engines, 24 APIs, 12 UI surfaces.

Phase 06 - AI Construction Intelligence: 120 requirements, 120 tests, 40 entities, 0 engines, 24 APIs, 12 UI surfaces.

Phase 07 - Enterprise Project Management: 160 requirements, 160 tests, 60 entities, 20 engines, 24 APIs, 12 UI surfaces.

Phase 08 - Enterprise CRM Sales Customer Experience: 180 requirements, 180 tests, 72 entities, 20 engines, 24 APIs, 12 UI surfaces.

Phase 09 - HR Workforce Talent Management: 210 requirements, 210 tests, 94 entities, 24 engines, 28 APIs, 15 UI surfaces.

Phase 10 - Enterprise Financial Management Accounting: 240 requirements, 240 tests, 110 entities, 25 engines, 32 APIs, 18 UI surfaces.

Phase 11 - Enterprise Procurement Supply Chain Inventory: 260 requirements, 260 tests, 120 entities, 30 engines, 36 APIs, 22 UI surfaces.

Phase 12 - Enterprise Fleet Asset Maintenance Telematics: 280 requirements, 280 tests, 125 entities, 32 engines, 40 APIs, 22 UI surfaces.

Phase 13 - Enterprise Business Intelligence Decision Intelligence: 300 requirements, 300 tests, 123 entities, 35 engines, 45 APIs, 25 UI surfaces.

Phase 14 - Enterprise HSE Safety Environmental Quality Compliance: 320 requirements, 320 tests, 145 entities, 40 engines, 50 APIs, 35 UI surfaces.

Phase 15 - Enterprise Document Drawing Specification Information Management: 340 requirements, 340 tests, 155 entities, 45 engines, 54 APIs, 38 UI surfaces.

Phase 16 - GIS Survey Reality Capture Drone Operations: 360 requirements, 360 tests, 165 entities, 49 engines, 61 APIs, 42 UI surfaces.

Phase 17 - Machine Control GPS Field Automation: 380 requirements, 380 tests, 175 entities, 55 engines, 65 APIs, 49 UI surfaces.

Phase 18 - Enterprise Scheduling Resource Planning: 400 requirements, 400 tests, 185 entities, 59 engines, 63 APIs, 49 UI surfaces.

Phase 19 - Contract Change Order Claims Commercial Management: 420 requirements, 420 tests, 186 entities, 53 engines, 67 APIs, 49 UI surfaces.

Phase 20 - Enterprise Integration Deployment Testing Governance Production Readiness: 450 requirements, 450 tests, 207 entities, 66 engines, 83 APIs, 50 UI surfaces.

11. Structural Reconciliation

278 machine-detected structural findings are recorded in the complete Gap/Duplicate/Conflict Register. These findings require review according to severity and do not replace semantic cross-phase review.

12. Final Acceptance

After Phase 00-21 review, cross-phase reconciliation, approval and manifest verification, the locked GES v1.0 baseline becomes the authoritative specification for architecture, backlog decomposition, code generation, integration, testing, release and future controlled change.